

# Jeonghwan Kim

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## Research Interests

### Natural Language Processing (NLP), Multimodal Large Language Models (MLLMs), Deep Learning

Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), Multimodal Representation Learning, Multi-Hop/Open-Domain Question Answering (QA), Numeracy in LLMs, and Model Interpretability and Robustness.

## Education

### University of Illinois at Urbana-Champaign (UIUC)

Ph.D. Candidate in Computer Science (Advisor: [Heng Ji](#))

Champaign, USA  
Aug 2023 – Present

### KAIST

M.S. in Computer Science (Advisor: [Sung-Hyon Myaeng](#))  
Thesis Committee: Sung-Hyon Myaeng, Alice Oh, Junho Lim  
GPA: 4.13/4.30

Daejeon, South Korea  
Feb 2022

### Handong Global University

B.S. in Computer Science & Electrical Eng. (Advisor: [Heeyoul Henry Choi](#))  
Magna Cum Laude (Overall GPA: 4.11/4.5 | Major GPA: 4.21/4.5)

Pohang, South Korea  
Feb 2020

## Experience

### Meta — Research Scientist Intern

Working on multimodal LLMs.

Redmond, USA  
May 2026 – Present

### Meta — Research Scientist Intern (Part-time Student Researcher)

Working on Retrieval-Augmented Generation (RAG) on multimodal LLMs.

Redmond, USA  
May 2025 – Oct 2025

### Amazon — Applied Scientist Intern

Developed a generalizable multimodal encoder for geospatial applications.

Bellevue, USA  
May 2024 – Aug 2024

### UIUC — Research Assistant

Research on Multimodal LLMs and reasoning.

Champaign, USA  
Aug 2023 – Present

### KAIST — Researcher

Developed graph and RAG-based language modeling approaches for QA.

Daejeon, Korea  
Mar 2022 – Jul 2023

### Republic of Korea Marine Corps — Sergeant (Honorably Discharged)

Served as a rifleman and interpreter for the ROK-US Combined Marine Command.

Pohang, Korea  
Mar 2015 – Dec 2016

## Publications

‘\*’ indicates equal contribution

### [1] Pixel-Grounded Retrieval for Knowledgeable Large Multimodal Models

**Jeonghwan Kim**, Renjie Tao, Sanat Sharma, Jiaqi Wang, Kai Sun, Zhaojiang Lin, Seungwhan Moon, Lambert Mathias, Anuj Kumar, Heng Ji, Xin Luna Dong

Preprint (Under Review)

### [2] Training VLAs with Vision-Language Anchoring and Language-Action Alignment

Dwip Dalal, Shivansh Patel, **Jeonghwan Kim**, Utkarsh Mishra, Alex Baratian, Hyeonjeong Ha, Heng Ji, Svetlana Lazebnik, Unnat Jain

Preprint (Under Review)

- [3] **Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping**  
*Dwip Dalal, Gautam Vashishtha, Utkarsh Mishra, Jeonghwan Kim, Madhav Kanda, Hyeonjeong Ha, Svetlana Lazebnik, Heng Ji, Unnat Jain*  
**ICLR 2026**
- [4] **PEARL: Self-Evolving Assistant for Time Management with Reinforcement Learning**  
*Bingxuan Li, Jeonghwan Kim, Cheng Qian, Xiusi Chen, Eitan Anzenberg, Niranjana Kundapur, Heng Ji*  
**ACL 2026**
- [5] **MM-PoisonRAG: Disrupting Multimodal RAG with Local and Global Knowledge Poisoning Attacks**  
*Hyeonjeong Ha\*, Qiusi Zhan\*, Jeonghwan Kim, Dimitrios Bralios, Saikrishna sanniboina, Nanyun Peng, Kai-Wei Chang, Daniel Kang, Heng Ji*  
**ACL 2026**
- [6] **PARTONOMY: Large Multimodal Models with Part-Level Visual Understanding**  
*Jeonghwan Kim\*, Ansel Blume\*, Hyeonjeong Ha, Elen Chatikyan, Xiaomeng Jin, Khanh Duy Nguyen, Nanyun Peng, Kai-Wei Chang, Derek Hoiem and Heng Ji*  
**NeurIPS 2025 (Spotlight)**
- [7] **Infogent: An Agent-based Framework for Web Information Aggregation**  
*Jeonghwan Kim\*, Revanth Gangi Reddy\*, Sagnik Mukherjee\*, Zhenhailong Wang\*, Dilek Hakkani-Tur, Heng Ji*  
**NAACL 2025, Findings**
- [8] **SYNTHIA: Novel Concept Design with Affordance Composition**  
*Hyeonjeong Ha, Xiaomeng Jin, Jeonghwan Kim, Jiateng Liu, Zhenhailong Wang, Khanh Duy Nguyen, Ansel Blume, Nanyun Peng, Kai-Wei Chang, Heng Ji*  
**ACL 2025**
- [9] **Search and Detect: Training-Free Long Tail Object Detection via Web-Image Retrieval**  
*Mankeerat Sidhu, Hetarth Chopra, Ansel Blume, Jeonghwan Kim, Revanth Gangi Reddy, Heng Ji*  
**CVPR 2025**
- [10] **Aligning LLMs with Individual Preferences via Interaction**  
*Shujin Wu, May Fung, Cheng Qian, Jeonghwan Kim, Dilek Hakkani-Tur, Heng Ji*  
**COLING 2025**
- [11] **Finer: Investigating and Enhancing Fine-Grained Visual Concept Recognition in Large Vision Language Models**  
*Jeonghwan Kim, Heng Ji*  
**EMNLP 2024**
- [12] **Why So Gullible? Enhancing the Robustness of Retrieval-Augmented Models against Counterfactual Noise**  
*Giwon Hong\*, Jeonghwan Kim\*, Junmo Kang\*, Sung-Hyon Myaeng, Joyce Jiyoung Whang*  
**NAACL 2024, Findings**
- [13] **ARMADA: Attribute-Based Multimodal Data Augmentation**  
*Xiaomeng Jin, Jeonghwan Kim, Yu Zhou, Kuan-Hao Huang, Te-Lin Wu, Nanyun Peng, Heng Ji*  
**Advancing Natural Language Process for Wikipedia@EMNLP 2024**
- [14] **MIRACLE: An Online, Explainable Multimodal Interactive Concept Learning System**  
*Ansel Blume\*, Khanh Duy Nguyen\*, Zhenhailong Wang, Yangyi Chen, Michal Shlapentokh-Rothman, Xiaomeng Jin, Jeonghwan Kim, Zhen Zhu, Jiateng Liu, Kuan-Hao Huang, Mankeerat Sidhu, Xuanming Zhang, Vivian Liu, Raunak Sinha, Te-Lin Wu, Abhay Zala, Elias Stengel-Eskin, Da Yin, Yao Xiao, Utkarsh Mall, Zhou Yu, Kai-Wei Chang, Camille Cobb, Karrie Karahalios, Lydia Chilton, Mohit Bansal, Nanyun Peng, Carl Vondrick, Derek Hoiem, Heng Ji*  
**ACM MM Technical Demos, 2024**
- [15] **FinePrompt: Unveiling the Role of Finetuned Inductive Bias on Compositional Reasoning in GPT-4**  
*Jeonghwan Kim\*, Giwon Hong\*, Sung-Hyon Myaeng, Joyce Jiyoung Whang*  
**EMNLP 2023, Findings**
- [16] **Graph-Induced Transformers for Efficient Multi-Hop Question Answering**  
*Giwon Hong, Jeonghwan Kim, Junmo Kang, Sung-Hyon Myaeng*  
**EMNLP 2022**

- [17] **Exploiting Numerical-Contextual Knowledge to Improve Numerical Reasoning in Question Answering**  
*Jeonghwan Kim, Junmo Kang, Kyung-min Kim, Giwon Hong, Sung-Hyon Myaeng*  
**NAACL 2022, Findings**
- [18] **Have You Seen That Number? Investigating Extrapolation in Question Answering Models**  
*Jeonghwan Kim, Giwon Hong, Kyung-min Kim, Junmo Kang, Sung-Hyon Myaeng*  
**EMNLP 2021**
- [19] **Leveraging Order-Free Tag Relations for Context-Aware Recommendation**  
*Junmo Kang, Jeonghwan Kim, Suwon Shin, Sung-Hyon Myaeng*  
**EMNLP 2021**
- [20] **Can You Distinguish Truthful from Fake Reviews? User Analysis and Assistance Tool for Fake Review Detection**  
*Jeonghwan Kim\*, Junmo Kang\*, Suwon Shin\*, Sung-Hyon Myaeng*  
**HCI+NLP Workshop, EACL 2021**
- [21] **Maximizing Efficiency of Language Model Pre-training for Learning Representation**  
*Junmo Kang\*, Suwon Shin\*, Jeonghwan Kim\*, Jaeyoung Jo\*, Sung-Hyon Myaeng*  
**KSC 2020**
- [22] **Object Classification on Raw Radar Data Using Convolutional Neural Networks**  
*Heejae Han, Jeonghwan Kim, Junyoung Park, YuJin Lee, Hyunwoo Jo, Yonghyeon Park, Eric T Matson, Seongha Park*  
**IEEE SAS 2019**
- [23] **Towards the Development and Realization of an Undetectable Stealth UAV**  
*Jiyeon Oh, Daeun Choe, Chanhui Yun, Jeonghwan Kim, Michael Hopmeier*  
**IEEE IRC 2019**

## Academic Services

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**Reviewer** - NeurIPS 2026, ICML 2026, ECCV 2026, CVPR 2026, ACL 2026, EACL 2026, SEA@NeurIPS 2025, AI4Research@AAAI 2025, EMNLP 2025, ACL 2025, ACL-SRW 2025, AAAI 2025, ACL 2024, Language + Molecules@ACL 2024, KnowledgeLM@ACL 2024, EMNLP 2024, EMNLP 2023

## Projects

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>DARPA Environment-driven Conceptual Learning (ECOLE)</b><br><i>Funded by DARPA</i>                                                                                                                   | Aug 2023 – Present  |
| <b>DARPA In the Moment (ITM)</b><br><i>Funded by DARPA</i>                                                                                                                                              | Aug 2023 – Present  |
| <b>Center for Advanced Bioenergy and Bioproducts Innovation (CABBI)</b><br><i>Funded by U.S. Department of Energy</i>                                                                                   | Aug 2024 – Present  |
| <b>Development of Expert Decision-making Support AI</b><br>Development of AI technology for explainable expert decision-making.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i> | Apr 2022 – Jul 2022 |
| <b>Contextual-Numerical Embedding for Reasoning</b><br>Development of context-number embedding for numerical reasoning in QA.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>   | Jul 2021 – Dec 2021 |
| <b>ExoBrain</b><br>Development of Knowledge Evolutionary WiseQA Platform Technology.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>                                            | Apr 2020 – Jul 2023 |
| <b>Smart Interaction and Context Association</b><br>Machine Learning for Context Association and Smart Interaction Suggestion.<br><i>Funded by the Korean Government (Ministry of Science and ICT)</i>  | Apr 2020 – Mar 2021 |

**AI-based National Online Petition System**

Development of AI-based national Online Petition System for citizen deliberation.  
*Funded by KAISTs Exploratory Research Program*

Apr 2021 – Dec 2021

**Purdue University - IITP Capstone**

Raw radar data object detection and stealth UAV design.  
*Funded by the Korean Government (IITP)*

Aug 2018 – Dec 2018

## Honors & Awards

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- **2026:** Great Reviewer - ACL Rolling Review, January, 2026 (Association for Computational Linguistics)
- **2020:** Graduated with Honors in CSE, Magna Cum Laude (Handong Global University)
- **2020:** Finalist (Top 3), COMEUP2020 AI CHAMPIONSHIP (K-Startup)
- **2019:** Grand Prize, Capstone Graduation Project Festival (Handong Global University, CSEE)
- **2018:** Certificate of Merit, IITP Capstone 9 Program (Purdue University)
- **2018:** Finalist (Top 15), NAVER AI Hackathon (NAVER AI RUSH)
- **2017:** National Science and Technology Scholarship, Korea Student Aid Foundation (KOSAF)

## Teaching

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- **Aug 2020 – Jun 2021:** Teaching Assistant, Senior Data Science, Samsung SDS
- **Jun 2021 – Jul 2021:** Teaching Assistant, Text Mining, SKKU Global Business School
- **Mar 2021 – Dec 2021:** Teaching Assistant, Text Mining, KAIST
- **Mar 2019 – Jun 2019:** Teaching Assistant, Data Structures, Handong Global University

## Technical Skills

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**Programming:** Python, C, C++, Java

**Frameworks:** Pytorch, TensorFlow, Theano, Docker, Huggingface, OpenCV, Flutter, Firebase, DGL

**Tools:** SpaCy, NLTK, Seaborn, Matplotlib